

Fourier method for PDE.

A piecewise smooth function $f(x)$ defined on the interval $0 \leq x \leq l$, can be expanded in the Fourier cosine or Fourier sine series,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l}, \quad a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx, \quad (1)$$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}, \quad b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx. \quad (2)$$

Examples

1. A string of length l fixed at the ends and initially at rest, is struck at the point $x = a$, so that momentum P is imparted to it. Determine the motion of the string.

Hint: solve the one-dimensional wave equation with the initial conditions

$$u(x, 0) = 0, \quad u_t(x, 0) = \begin{cases} 0, & 0 < x < a - \varepsilon, \\ v_0, & a - \varepsilon < x < a + \varepsilon, \\ 0, & a + \varepsilon < x < l, \end{cases} \quad (3)$$

where $v_0 = P/(2\varepsilon\rho)$ is the initial velocity over a small interval 2ε near $x = a$, and ρ is the linear mass density. Then take the limit $\varepsilon \rightarrow 0$.

2. Find the temperature $u(x, t)$ of the rod of length l , whose ends are insulated [i.e., no heat flux, $u_x(0, t) = u_x(l, t) = 0$], given the initial temperature distribution $u(x, 0) = f(x)$.
3. Determine the vibrations of a string with fixed ends, initially at rest, produced by a load with constant density q applied at $t = 0$ over the whole length of the string.

[Hint: seek solution in the form $u(x, t) = \tilde{u}(x, t) + u_s(x)$, where $u_s(x)$ is the stationary solution of the wave equation with the load, $u_{tt} - c^2 u_{xx} = q/\rho$.]

Homework problems

1. Find the temperature of the rod $u(x, t)$, whose ends are kept at zero temperature, given that its initial temperature is T , by solving the heat equation

$$u_t - K u_{xx} = 0, \quad (4)$$

with boundary conditions $u(0, t) = u(l, t) = 0$ and initial condition $u(x, 0) = T$. [20]

Answer:
$$u(x, t) = \frac{4T}{\pi} \sum_{m=0}^{\infty} \frac{\sin \frac{(2m+1)\pi x}{l}}{2m+1} e^{-[(2m+1)^2 \pi^2 / l^2] K t}.$$

2. Solve the same problem for the parabolic initial temperature distribution, [10]

$$u(x, 0) = \frac{4T}{l^2} x(l - x).$$

Answer:
$$u(x, t) = \frac{32T}{\pi^3} \sum_{m=0}^{\infty} \frac{\sin \frac{(2m+1)\pi x}{l}}{(2m+1)^3} e^{-[(2m+1)^2 \pi^2 / l^2] K t}.$$

3. Solve the heat equation for the rod with insulating ends, $u_x(0, t) = u_x(l, t) = 0$, and initial temperature distribution [20]

$$u(x, 0) = \frac{4T}{l^2}x(l - x).$$

Answer:
$$u(x, t) = \frac{2T}{3} - \frac{16T}{\pi^2} \sum_{m=1}^{\infty} \frac{\cos \frac{2m\pi x}{l}}{4m^2} e^{-[4m^2\pi^2/l^2]Kt}.$$

[Note that unlike questions 1 and 2, the temperature here does not tend to zero for $t \rightarrow \infty$, since the heat energy cannot escape the rod.]

4. Determine the displacement of the string, $u(x, t)$, with fixed ends, $u(0, t) = u(l, t) = 0$, given that it is initially at rest and has a parabolic shape with maximum displacement h , [20]

$$u(x, 0) = \frac{4h}{l^2}x(l - x), \quad u_t(x, 0) = 0.$$

Answer:
$$u(x, t) = \frac{32h}{\pi^3} \sum_{m=0}^{\infty} \frac{\sin \frac{(2m+1)\pi x}{l}}{(2m+1)^3} \cos \frac{(2m+1)\pi c}{l}t.$$

5. A rectangular membrane of sides $2a$ and $2b$, whose edge is fixed, is acted upon by a uniformly distributed load q . The equilibrium shape of the membrane, $u(x, y)$, obeys the equation

$$u_{xx} + u_{yy} = -\frac{q}{T}, \quad (5)$$

where T is the tension per unit length in the membrane, $-a \leq x \leq a$, $-b \leq y \leq b$, and

$$u(-a, y) = u(a, y) = u(x, -b) = u(x, b) = 0. \quad (6)$$

- (a) Seek solution of Eq. (5) in the form $u_1(x)$ with boundary conditions $u_1(-a) = 0$, $u_1(a) = 0$, and show that [5]

$$u_1(x) = \frac{q}{2T}(a^2 - x^2). \quad (7)$$

- (b) Let $u(x, y) = \tilde{u}(x, y) + u_1(x)$. By substitution into equation (5) and conditions (6), show that $\tilde{u}(x, y)$ satisfies the homogeneous equation [5]

$$\tilde{u}_{xx} + \tilde{u}_{yy} = 0, \quad (8)$$

with boundary conditions $\tilde{u}(-a, y) = \tilde{u}(a, y) = 0$, $\tilde{u}(x, -b) = \tilde{u}(x, b) = -u_1(x)$.

- (c) Seek solution of (8) as $\tilde{u}(x, y) = X(x)Y(y)$, with $X(-a) = X(a) = 0$, and show that

$$\tilde{u}(x, y) = A_n \cosh \frac{(n + \frac{1}{2})\pi y}{a} \cos \frac{(n + \frac{1}{2})\pi x}{a}, \quad (9)$$

where A_n is an arbitrary constant, and $n = 0, 1, 2, \dots$ [10]

- (d) Using the superposition principle, construct a more general $\tilde{u}(x, y)$ as a sum of (9). By subjecting it to the remaining boundary conditions, $\tilde{u}(x, -b) = \tilde{u}(x, b) = -u_1(x)$, find A_n , and show that [10]

$$u(x, y) = \frac{qa^2}{2T} \left[1 - \frac{x^2}{a^2} + \frac{32}{\pi^3} \sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{(2n+1)^3} \frac{\cosh \frac{(2n+1)\pi y}{2a}}{\cosh \frac{(2n+1)\pi b}{2a}} \cos \frac{(2n+1)\pi x}{2a} \right].$$